

Strategy 1: Always begin by attempting to “find” the conclusion in the premises.

Strategy 2: If the conclusion contains a letter that appears in the consequent of a conditional statement in the premises, consider obtaining that letter via *modus ponens*:

1. $A \supset B$
2. $C \vee A$
3. A // B
4. B 1, 3, MP

Strategy 3: If the conclusion contains a negated letter and that letter appears in the antecedent of a conditional statement in the premises, consider obtaining the negated letter via *modus tollens*:

1. $C \supset B$
2. $A \supset B$
3. $\sim B$ // $\sim A$
4. $\sim A$ 2, 3, MT

Strategy 4: If the conclusion is a conditional statement, consider obtaining it via pure hypothetical syllogism:

1. $B \supset C$
2. $C \supset A$
3. $A \supset B$ // $A \supset C$
4. $A \supset C$ 1, 3, HS

Strategy 5: If the conclusion contains a letter that appears in a disjunctive statement in the premises, consider obtaining that letter via disjunctive syllogism:

1. $A \supset B$
2. $A \vee C$
3. $\sim A$ // C
4. C 2, 3, DS

Of course, these strategies apply to deriving any line prior to the conclusion, just as they apply to deriving the conclusion.

Strategy 6: If the conclusion contains a letter that appears in a conjunctive statement in the premises, consider obtaining that letter via simplification:

1. $A \supset B$
2. $C \cdot B$
3. $C \supset A$ // C
4. C 2, Simp

Strategy 7: If the conclusion is a conjunctive statement, consider obtaining it via conjunction by first obtaining the individual conjuncts:

1. $A \supset C$
2. B
3. $\sim C$ $// B \cdot \sim C$
4. $B \cdot \sim C$ 2, 3, Conj

Strategy 8: If the conclusion is a disjunctive statement, consider obtaining it via constructive dilemma or addition:

1. $(A \supset B) \cdot (C \supset D)$
2. $B \supset C$
3. $A \vee C$ $// B \vee D$
4. $B \vee D$ 1, 3, CD

1. $A \vee C$
2. B
3. $C \supset D$ $// B \vee D$
4. $B \vee D$ 2, Add

Strategy 9: If the conclusion contains a letter not found in the premises, addition *must* be used to introduce that letter.

Strategy 10: Conjunction can be used to set up constructive dilemma:

1. $A \supset B$
2. $C \supset D$
3. $A \vee C$ $// B \vee D$
4. $(A \supset B) \cdot (C \supset D)$ 1, 2, Conj
5. $B \vee D$ 3, 4, CD

Strategy 11: Conjunction can be used to set up De Morgan's rule:

1. $\sim A$
2. $\sim B$
3. $\sim A \cdot \sim B$ 1, 2, Conj
4. $\sim(A \vee B)$ 3, DM

Strategy 12: Constructive dilemma can be used to set up De Morgan's rule:

1. $(A \supset \sim B) \cdot (C \supset \sim D)$
2. $A \vee C$
3. $\sim B \vee \sim D$ 1, 2, CD
4. $\sim(B \cdot D)$ 3, DM

Strategy 13: Addition can be used to set up De Morgan's rule:

1. $\sim A$
2. $\sim A \vee \sim B$ 1, Add
3. $\sim(A \cdot B)$ 2, DM

Strategy 14: Distribution can be used in two ways to set up disjunctive syllogism:

1. $(A \vee B) \cdot (A \vee C)$
2. $\sim A$
3. $A \vee (B \cdot C)$ 1, Dist
4. $B \cdot C$ 2, 3, DS

1. $A \cdot (B \vee C)$
2. $\sim(A \cdot B)$
3. $(A \cdot B) \vee (A \cdot C)$ 1, Dist
4. $A \cdot C$ 2, 3, DS

Strategy 15: Distribution can be used in two ways to set up simplification:

1. $(A \cdot B) \vee (A \cdot C)$
2. $A \cdot (B \vee C)$ 1, Dist
3. A 2, Simp

Strategy 16: If inspection of the premises does not reveal how the conclusion should be derived, consider using the rules of replacement to deconstruct the conclusion. (See the final example in this section.)

Strategy 17: Material implication can be used to set up hypothetical syllogism:

1. $\sim A \vee B$
2. $\sim B \vee C$
3. $A \supset B$ 1, Impl
4. $B \supset C$ 2, Impl
5. $A \supset C$ 3, 4, HS

Strategy 18: Exportation can be used to set up *modus ponens*:

1. $(A \cdot B) \supset C$
2. A
3. $A \supset (B \supset C)$ 1, Exp
4. $B \supset C$ 2, 3, MP

Strategy 19: Exportation can be used to set up *modus tollens*:

1. $A \supset (B \supset C)$
2. $\sim C$
3. $(A \cdot B) \supset C$ 1, Exp
4. $\sim(A \cdot B)$ 2, 3, MT

Strategy 20: Addition can be used to set up material implication:

1. A
2. $A \vee \sim B$ 1, Add
3. $\sim B \vee A$ 2, Com
4. $B \supset A$ 3, Impl

Strategy 21: Transposition can be used to set up hypothetical syllogism:

1. $A \supset B$
2. $\sim C \supset \sim B$
3. $B \supset C$ 2, Trans
4. $A \supset C$ 1, 3, HS

Strategy 22: Transposition can be used to set up constructive dilemma:

1. $(A \supset B) \cdot (C \supset D)$
2. $\sim B \vee \sim D$
3. $(\sim B \supset \sim A) \cdot (C \supset D)$ 1, Trans
4. $(\sim B \supset \sim A) \cdot (\sim D \supset \sim C)$ 3, Trans
5. $\sim A \vee \sim C$ 2, 4, CD

Strategy 23: Constructive dilemma can be used to set up tautology:

1. $(A \supset C) \cdot (B \supset C)$
2. $A \vee B$
3. $C \vee C$ 1, 2, CD
4. C 3, Taut

Strategy 24: Material implication can be used to set up tautology:

1. $A \supset \sim A$
2. $\sim A \vee \sim A$ 1, Impl
3. $\sim A$ 2, Taut

Strategy 25: Material implication can be used to set up distribution:

1. $A \supset (B \cdot C)$
2. $\sim A \vee (B \cdot C)$ 1, Impl
3. $(\sim A \vee B) \cdot (\sim A \vee C)$ 2, Dist

From Hurley's "A concise introduction to Logic" Chapter 7